

QUESTION:6 WRITE A C PROGRAM FOR INSERTION OF A NEW NODE AT THE END OF A SINGLE LINKED LIST

main.c	Output
<pre>1 #include <stdio.h> 2 #include <stdlib.h> 3 struct Node { 4 int data; 5 struct Node *next; 6 }; 7 struct Node* insertAtEnd(struct Node *head, int value) { 8 struct Node *newNode, *temp; 9 newNode = (struct Node*)malloc(sizeof(struct Node)); 10 newNode->data = value; 11 newNode->next = NULL; 12 if (head == NULL) { 13 return newNode; 14 } 15 temp = head; 16 while (temp->next != NULL) { 17 temp = temp->next; 18 } 19 temp->next = newNode; 20 21 return head; 22 }</pre>	<pre>Linked List before insertion: 10 -> 20 -> 30 -> NULL Linked List after insertion at the end: 10 -> 20 -> 30 -> 40 -> NULL === Code Execution Successful ===</pre>
<pre>23 void display(struct Node *head) { 24 struct Node *temp = head; 25 while (temp != NULL) { 26 printf("%d -> ", temp->data); 27 temp = temp->next; 28 } 29 printf("NULL\n"); 30 } 31 int main() { 32 struct Node *head = NULL; 33 head = insertAtEnd(head, 10); 34 head = insertAtEnd(head, 20); 35 head = insertAtEnd(head, 30); 36 printf("Linked List before insertion:\n"); 37 display(head); 38 head = insertAtEnd(head, 40); 39 printf("Linked List after insertion at the end:\n"); 40 display(head); 41 return 0; 42 }</pre>	